

How AI Will Affect Cybersecurity

Artificial intelligence (AI) is becoming an important part of cybersecurity and will continue to change how organizations protect their systems and data. AI can help cybersecurity professionals analyze large amounts of information, identify unusual activity, and respond to threats more efficiently. At the same time, attackers can also use AI to make cyberattacks more effective. Because of this, AI will likely create both new opportunities and new risks for cybersecurity professionals.

One of the biggest benefits of AI is its ability to process large amounts of data quickly. Security teams can use AI to analyze network traffic, security logs, alerts, and other information to identify patterns that may indicate an attack. AI can also help with threat detection and threat hunting by identifying activity that might otherwise be difficult for a human analyst to notice. NIST explains that AI has the potential to improve defensive cybersecurity capabilities, although organizations still need to manage issues such as false positives and the need for human oversight.

AI will also affect cybersecurity by changing the way attackers operate. AI-enabled tools can lower the amount of effort needed to create convincing phishing messages, automate certain activities, and adapt attacks more quickly. NIST has identified AI-related risks that include lowering barriers to cybersecurity attacks and has emphasized that organizations need to prepare for attackers who use AI as part of their methods.

Another major concern is the security of AI systems themselves. AI models depend on software, hardware, training data, and the information they receive. Attackers may attempt to manipulate that information or interfere with how an AI system makes decisions. NIST identifies several types of adversarial machine-learning attacks, including evasion, poisoning, privacy, and abuse attacks. These risks show that organizations will need to protect AI systems just like they protect other important technology.

Because AI can be used by both defenders and attackers, cybersecurity professionals will need to develop new skills. Professionals will need to understand how AI tools work, how to evaluate their results, and how to recognize AI-related security risks. Human judgment will remain important because AI systems can make mistakes, produce inaccurate results, or be manipulated. NIST recommends managing AI risks through structured approaches such as the AI Risk Management Framework and emphasizes the importance of secure and resilient AI systems.

Overall, AI will have a major impact on cybersecurity. It can help organizations detect threats faster, analyze information more effectively, and improve security operations. However, it can also give attackers new capabilities and create new vulnerabilities that organizations must

manage. The future of cybersecurity will likely involve humans and AI working together, with AI assisting security professionals while people provide oversight, judgment, and accountability.